Ps 04

Erik An

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1 Problem 1 DONE

1.1 Task

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x, y) = x^2 + y^2$.

(a) Compute $\nabla f(x, y)$.

(b) Evaluate it at $(1, 2)$ and determine the local (affine) linear approximation of f in $(1, 2)$.

(c) Plot f and ∇f in Julia.

1.2 Solution

1.2.1 (a)

Let $f : \mathbb{R}^2 \to \mathbb{R}, \quad f(x, y) = x^2 + y^2$.

By definition, the partial derivatives are:

$$\begin{aligned}
\partial_1 f(x, y) &= \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}, \\
\partial_2 f(x, y) &= \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}.
\end{aligned}$$

Computing $\partial_1 f$:

$$\begin{aligned}
\partial_1 f(x, y) &= \lim_{h \to 0} \frac{(x + h)^2 + y^2 - x^2 - y^2}{h} \\
&= \lim_{h \to 0} \frac{2xh + h^2}{h} \\
&= \lim_{h \to 0} (2x + h) = 2x.
\end{aligned}$$

By analogous computation, $\partial_2 f(x, y) = 2y$.

Therefore:

$$\nabla f(x, y) = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

1.2.2 (b)

Evaluating the gradient at $p = (1, 2)$:

$$\nabla f(1,2) = \begin{pmatrix} 2 \cdot 1 \\ 2 \cdot 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}.$$

The Fréchet derivative $Df(p) : \mathbb{R}^n \to \mathbb{R}^m$ is the unique linear map satisfying:

$$\lim_{h \to 0} \frac{\|f(p+h) - f(p) - Df(p)(h)\|}{\|h\|} = 0.$$

For $f : \mathbb{R}^n \to \mathbb{R}$, this linear functional has the representation $Df(p)(h) = \nabla f(p)^\top h$, yielding the affine approximation:

$$L(x) = f(p) + \nabla f(p)^\top (x - p)$$

With $p = (1, 2)$ and $f(1, 2) = 1^2 + 2^2 = 5$:

$$\begin{aligned} L(x,y) &= 5 + \begin{pmatrix} 2 & 4 \end{pmatrix} \begin{pmatrix} x-1 \\ y-2 \end{pmatrix} \\ &= 5 + 2(x-1) + 4(y-2) \\ &= 2x + 4y - 5. \end{aligned}$$

1.2.3 (c)

```
using Plots

f(x, y) = x^2 + y^2
L(x, y) = 2x + 4y - 5

xs = -6:0.05:6
ys = -6:0.05:6

anim = Animation()

for azimuth in 0:2:360
    surface(xs, ys, (x, y) -> f(x, y),
            alpha=0.8,
            colorbar=false,
            label="f(x,y)")
    surface!(xs, ys, (x, y) -> L(x, y),
```

```
            alpha=0.6,
            color=:reds,
            label="L(x,y)")
    scatter!([1], [2], [5],
            color=:black,
            markersize=4,
            label="(1,2,5)")
    plot!(xlabel="x", ylabel="y", zlabel="z",
          camera=(azimuth, 20),
          size=(800, 800))
    frame(anim)
end

gif(anim, "./assets/p_01_c.gif", fps=30)
```

```
Plots.AnimatedGif("/Users/ziro2173/personal/university/01_bachelor/02_semester/01_analysis/02_ex
```

2 Problem 2 DONE

2.1 Task

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be defined by $f(x, y, z) = xyz$.

(a) Compute $\nabla f(x, y, z)$.

(b) Evaluate it at $(1, 1, 1)$ and determine the local (affine) linear approximation of f in $(1, 1, 1)$.

(c) Plot $(x, y) \mapsto f(x, y, 1)$ in Julia.

2.2 Solution

2.2.1 (a)

Let $f : \mathbb{R}^3 \to \mathbb{R}, \quad f(x, y, z) = xyz$.

By definition, the partial derivatives are:

$$
\begin{aligned}
\partial_1 f(x,y,z) &= \lim_{h\to 0} \frac{f((x+h),y,z) - f(x,y,z)}{h} \\
\partial_2 f(x,y,z) &= \lim_{h\to 0} \frac{f(x,(y+h),z) - f(x,y,z)}{h} \\
\partial_3 f(x,y,z) &= \lim_{h\to 0} \frac{f(x,y,(z+h)) - f(x,y,z)}{h}
\end{aligned}
$$

Computing $\partial_1 f(x,y,z)$:

$$
\begin{aligned}
\partial_1 f(x,y,z) &= \lim_{h\to 0} \frac{f((x+h),y,z) - f(x,y,z)}{h} \\
&= \lim_{h\to 0} \frac{(x+h)yz - xyz}{h} \\
&= \lim_{h\to 0} \frac{hyz}{h} \\
&= yz
\end{aligned}
$$

By analogous computation, $\partial_2 f(x,y,z) = xz$, $\partial_3 f(x,y,z) = xy$.

Therefore:

$$
\nabla f(x,y,z) = \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}
$$

2.2.2 (b)

Evaluating the gradient at $p = (1,1,1)$:

$$
\nabla f(1,1,1) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}
$$

The Fréchet derivative $Df(p) : \mathbb{R}^3 \to \mathbb{R}$ is the unique linear map satisfying:

$$
\lim_{h\to 0} \frac{\|f(p+h) - f(p) - Df(p)(h)\|}{\|h\|} = 0
$$

For $f : \mathbb{R}^3 \to \mathbb{R}$, this linear functional has the representation $Df(p)(h) = \nabla f(p)^\top h$, yielding the affine approximation:

$$L(x) = f(p) + \nabla f(p)^\top (x - p)$$

With $p = (1, 1, 1)$, and $f(1, 1, 1) = 1$:

$$\begin{aligned} L(x, y, z) &= 1 + \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x - 1 \\ y - 1 \\ z - 1 \end{pmatrix} \\ &= 1 + (x - 1) + (y - 1) + (z - 1) \\ &= x + y + z - 2 \end{aligned}$$

2.2.3 (c)

```
using Plots

f(x, y) = x * y
L(x, y) = x + y - 1

xs = -6:0.05:6
ys = -6:0.05:6

anim = Animation()

for azimuth in 0:2:360
    surface(xs, ys, (x, y) -> f(x, y),
            alpha=0.8,
            colorbar=false,
            label="f(x,y)")
    surface!(xs, ys, (x, y) -> L(x, y),
             alpha=0.6,
             color=:reds,
             label="L(x,y)")
    scatter!([1], [1], [1],
             color=:black,
             markersize=4,
             label="(1,1,1)")
    plot!(xlabel="x", ylabel="y", zlabel="z",
          camera=(azimuth, 20),
          size=(800, 800))
    frame(anim)
```

```
end

gif(anim, "./assets/p_02_c.gif", fps=30)
```

```
Plots.AnimatedGif("/Users/ziro2173/personal/university/01_bachelor/02_semester/01_analysis/02_ex
```

3 Problem 3 DONE

3.1 Task

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x, y) = x^2y + y^2$.

(a) Compute $\operatorname{grad} f(x, y)$.

(b) Determine the local (affine) linear approximation of f at $(1, 1)$.

3.2 Solution

3.2.1 (a)

Let $f : \mathbb{R}^2 \to \mathbb{R}, \quad f(x, y) = x^2y + y^2$.

By definition, the partial derivatives are:

$$\begin{aligned}
\partial_1 f(x, y) &= \lim_{h \to 0} \frac{f((x + h), y) - f(x, y)}{h} \\
\partial_2 f(x, y) &= \lim_{h \to 0} \frac{f(x, (y + h)) - f(x, y)}{h}
\end{aligned}$$

Computing $\partial_1 f(x, y)$:

$$\begin{aligned}
\partial_1 f(x, y) &= \lim_{h \to 0} \frac{((x + h)^2y + y^2) - (x^2y + y^2)}{h} \\
&= \lim_{h \to 0} \frac{2xyh + yh^2}{h} \\
&= \lim_{h \to 0} 2xy + yh \\
&= 2xy
\end{aligned}$$

Computing $\partial_2 f(x,y)$:

$$\begin{aligned}
\partial_2 f(x,y) &= \lim_{h\to 0} \frac{(x^2(y+h)+(y+h)^2)-(x^2y+y^2)}{h} \\
&= \lim_{h\to 0} \frac{x^2h+2yh+h^2}{h} \\
&= \lim_{h\to 0} (x^2+2y+h) \\
&= x^2+2y
\end{aligned}$$

Therefore:

$$\nabla f(x,y) = \begin{pmatrix} 2xy \\ x^2+2y \end{pmatrix}$$

3.2.2 (b)

Evaluating the gradient at $p=(1,1)$:

$$\nabla f(1,1) = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

With $f(1,1) = 1+1 = 2$:

$$\begin{aligned}
L(x,y) &= 2 + \begin{pmatrix} 2 & 3 \end{pmatrix} \begin{pmatrix} x-1 \\ y-1 \end{pmatrix} \\
&= 2 + 2(x-1) + 3(y-1) \\
&= 2x+3y-3
\end{aligned}$$

4 Problem 4 DONE

4.1 Task

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x,y) = \frac{x}{x^2+y^2+1}$.

(a) Plot f in Julia.

(b) Guess its local extrema from your plot.

4.2 Solution

4.2.1 (a)

```
using Plots

f(x, y) = (x) / (x^2 + y^2 + 1)

xs = -10:0.1:10
ys = -10:0.1:10

anim = Animation()

for azimuth in 0:2:360
    surface(xs, ys, (x, y) -> f(x, y),
            alpha=0.8,
            color=:reds,
            label="f(x,y)")
    plot!(xlabel="x", ylabel="y", zlabel="z",
          camera=(azimuth, 20),
          size=(800, 800))
    frame(anim)
end

gif(anim, "./assets/p_04_a.gif", fps=30)
```

```
Plots.AnimatedGif("/Users/ziro2173/personal/university/01_bachelor/02_semester/01_analysis/02_ex
```

4.2.2 (b)

From the plot, f appears to have:

- a local maximum at $(1, 0)$ with $f(1, 0) = \frac{1}{2}$,
- a local minimum at $(-1, 0)$ with $f(-1, 0) = -\frac{1}{2}$.

5 Problem 5

5.1 Task

Let $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be twice continuously differentiable such that $f(x, y) = g(x + y)$.

(a) Show that the Taylor polynomial of g of order 1 centered at 0 and evaluated at $t = x + y$ coincides with the local (affine) linear approximation of f in $(0, 0)$.

(b) The Hessian matrix of f is

$$Hf(x, y) = \begin{pmatrix} \partial_x \partial_x f(x, y) & \partial_x \partial_y f(x, y) \\ \partial_y \partial_x f(x, y) & \partial_y \partial_y f(x, y) \end{pmatrix}.$$

Consider $g(t) = e^t$ and show that the Taylor polynomial of g of order 2 centered at 0 and evaluated at $t = x + y$ coincides with

$$F(x, y) = f(0, 0) + \operatorname{grad} f(0, 0) \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{2}(x, y)\, Hf(0, 0) \begin{pmatrix} x \\ y \end{pmatrix}.$$

5.2 Solution